

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 2 Theory

9189/2**JUNE 2018 SESSION**

1 hour 15 minutes

Candidates answer on the question paper.

Additional materials:

Data Booklet

Mathematical Tables and/or Calculator

TIME 1 hour 15 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
TOTAL	

This question paper consists of 10 printed pages and 2 blank pages.

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[Turn over

Answer all questions

For
Examiner's
Use

1 (a) State the acid /base nature of

1. carbon monoxide,

2. carbon dioxide.

[2]

(b) (i) Describe the structure, bonding and molecular shape of SiCl_4 .*structure*

bonding

molecular shape

(ii) Write a chemical equation for the hydrolysis of SiCl_4 .

(iii) State the observations made in the reaction in b(ii).

[5]

(c) State

(i) any **two** physical properties of ceramics,1.

2.

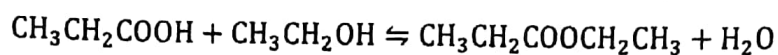
(ii) any **one** use of ceramics.

[3]

[Total:10]

2

At a given temperature, a mixture of propanoic acid and propanol reaches equilibrium according to the following equation:



$$\Delta H = -23 \text{ kJmol}^{-1}$$

When 7.40 g of propanoic acid, 9.20 g of ethanol and 9.00 g of water were made to reach equilibrium, the mixture contained 0.054 moles of ethyl propanoate.

- (a) (i) Calculate the number of moles of propanoic acid, ethanol and water in the equilibrium mixture

1. propanoic acid,

2. ethanol,

3. water.

[3]

- (ii) Calculate the value of K_c .

[3]

(b) Explain the effect of increasing temperature on the

(i) amount of the ester at equilibrium,

(ii) time taken to reach equilibrium.

[4]

[Total:10]

3 The reaction between calcium oxide and water is exothermic.

(a) (i) Write a balanced chemical equation for the reaction.

(ii) Table 3.1 shows standard enthalpies of formation of three compounds.

Table 3.1

compound	$\Delta H_f^\theta / \text{kJ mol}^{-1}$
CaO	-635
Ca(OH) ₂	-1003
H ₂ O	-286

Calculate the enthalpy change of the reaction in (i).

[3]

- (b) An experiment was set up as shown in Fig.3.1.

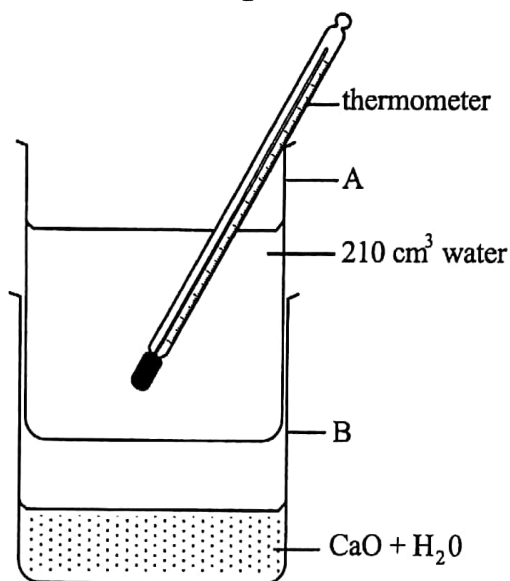


Fig.3.1

The temperature of the water in A rose by 40°C .

[Assume the specific heat capacity of the solution is $4,185 \text{ J K}^{-1} \text{ g}^{-1} \equiv 1 \text{ cm}^3$]

Calculate the

- (i) energy produced,
- (ii) minimum mass of calcium oxide needed to heat 210 cm^3 of water by 40°C .

[3]

(c) Some matchstick heads contain a mixture of tetraphosphorus trisulphide; P_4S_3 , and potassium chlorate (V), $KClO_3$. $KClO_3$ decomposes to give KCl and O_2 needed for the combustion of P_4S_3 .

(i) Write a balanced chemical equation for the

1. combustion of P_4S_3 ,
2. decomposition of $KClO_3$,
3. overall reaction that takes place when a match head ignites.

(ii) State the ratio in which P_4S_3 and $KClO_3$ should be combined for the match head to ignite.

[4]

[Total:10]

- 4 The reaction scheme shown in Fig.4.1 shows how compound A can be converted into organic compounds B, C and D.

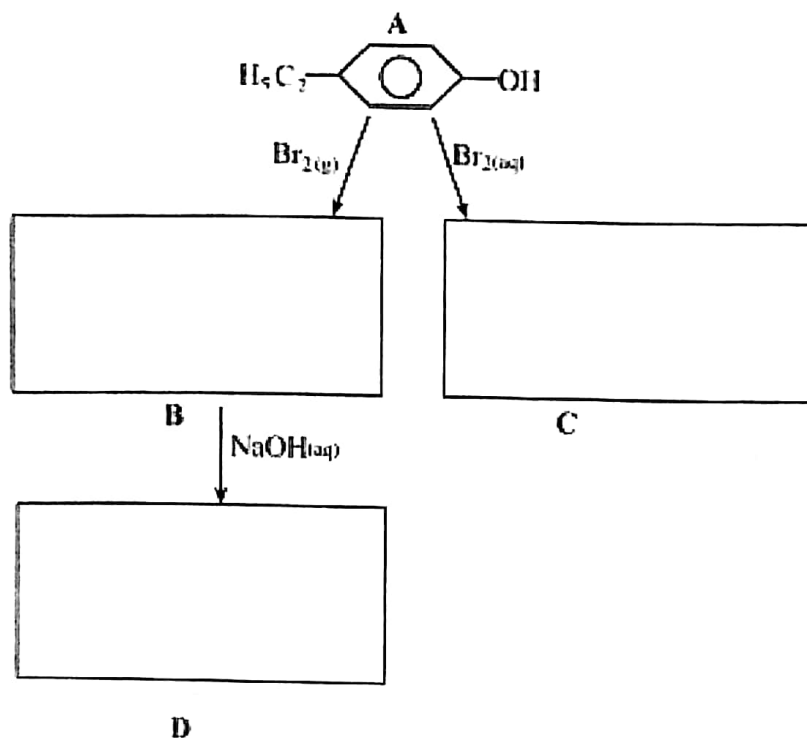


Fig.4.1

- (a) (i) Complete the reaction scheme by drawing the displayed structural formula of organic compounds B, C and D.
- (ii) Name the type of reaction that leads to formation of
1. **B**, _____
 2. **C**, _____
- (iii) Describe a chemical test that can be used to distinguish between C and D.

[7]

(b) (i) Describe the mechanism of the reaction that leads to the formation of **B**.

(ii) State the observable changes that occur during the formation of

1. **B**,

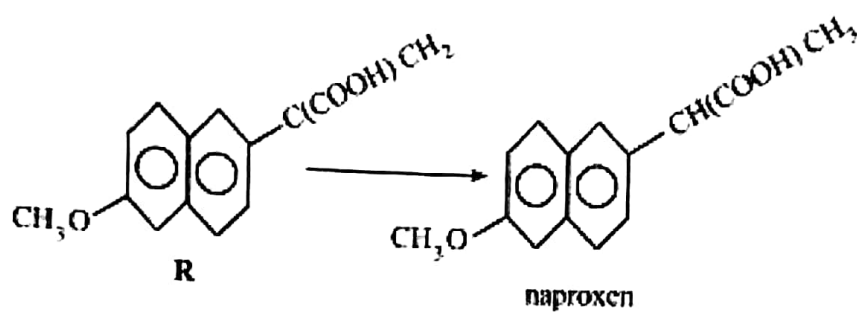
2. **C**.

[4]

[Total:11]

5

A drug, Naproxen, can be made from compound **R** as shown:



- (a) State the reagents and conditions for the conversion of **R** into naproxen.

reagents _____

conditions _____

[2]

- (b) (i) Draw the displayed formulae of the two isomeric forms of naproxen.

- (ii) State the type of isomerism shown.

[2]

(c) Draw the structural formula of the organic product(s) formed when compound R reacts with

(i) phenylamine,

(ii) hot concentrated acidified potassium permanganate,

(iii) phosphoric acid and water.

[3]
[Total:7]